

Name: _____

Date: April 29, 2025

Total
26.5
/34

1. Determine the derivative of the following functions. Simplify and factor where possible.

a) $f(x) = 10^{\sqrt{e^x - \sin x}}$

$$f(x) = 10^{(e^x - \sin x)^{\frac{1}{2}}}$$

$$f'(x) = 10^{\sqrt{e^x - \sin x}} (\ln 10) (e^x - \cos x) (e^x - \sin x)^{-\frac{1}{2}} \left(\frac{1}{2}\right)$$

$$= \frac{10^{\sqrt{e^x - \sin x}} (\ln 10) (e^x - \cos x)}{2 \sqrt{e^x - \sin x}}$$

c) $f(x) = \frac{\sin x}{e^{x^3}}$

$$f(x) = \frac{\sin x}{e^{x^3}}$$

$$f'(x) = \frac{(\cos x)(e^{x^3}) - (e^{x^3})(3x^2)(\sin x)}{(e^{x^3})^2}$$

$$= \frac{\cos x - 3x^2 \sin x}{e^{x^3}}$$

b) $y = \ln(\sin 4x + e^x) + 4^x$

$$y = \ln(\sin 4x + e^x) + 4^x$$

$$y' = \frac{4 \cos x + e^x}{\sin 4x + e^x} + 4^x (\ln 4) (\sin 4x + e^x)$$

$$= \frac{4 \cos x + e^x}{\sin 4x + e^x} + 4^x (\ln 4) (\sin 4x + e^x)$$

d) $y = \sqrt[2]{2x^6 \cos x} + \sqrt[3]{3x^7 \sin x}$

$$y = 2x^6 \cos x + 3x^7 \sin x$$

$$y' = [12x^5 \cos x + 2x^6 (-\sin x)] + [21x^6 \sin x + 3x^7 \cos x]$$

$$= 12x^5 \cos x + 19x^6 \sin x + 3x^7 \cos x$$

Factor GCF.

2. Determine the equation of the tangent to the curve $y = \log_4(4x^2 - 5x)$ at $x = 2$. Use exact values.

$$y = \log_4(4x^2 - 5x)$$

$$y' = \frac{8x - 5}{(4x^2 - 5x) \ln 4}$$

$$y'(2) = \frac{8(2) - 5}{(4(2)^2 - 5(2)) \ln 4}$$

$$= \frac{11}{6 \ln 4} \rightarrow m$$

$$P(2, \log_4 6)$$

$$y = mx + b$$

$$\log_4 6 = \frac{11}{6 \ln 4} (2) + b$$

$$\log_4 6 - \frac{11}{3 \ln 4} = b$$

$$y = \frac{11}{6 \ln 4} (x) + \left(\log_4 6 - \frac{11}{3 \ln 4} \right)$$

10.5

$$[-2\pi, 2\pi]$$

3. Find all x-values on the graph of $f(x) = x - 2\cos x$ for $[-2\pi, 2\pi]$ where the tangent line is normal to $y = \frac{1}{2}$.

$\frac{1}{2}$ slope $\rightarrow 2 = \text{normal}$

$$f(x) = x - 2\cos x$$

$$f'(x) = 1 + 2\sin x$$

$$2 = 1 + 2\sin x$$

$$\frac{1}{2} = \sin x$$

$$x = \sin^{-1}\left(\frac{1}{2}\right)$$

$$= \frac{\pi}{6}$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6}, -\frac{7\pi}{6}, -\frac{11\pi}{6}$$

4. The spread of a rumor in a certain school is modeled by the equation $P(t) = \frac{300}{1+2^{4-t}}$ where $P(t)$ is the total number of students who have heard the rumor t days after the rumor first started to spread.

- a) Approximately, what is the initial number of students who heard the rumor?

$$P(0) = \frac{300}{1+2^{4-0}} = \frac{300}{17} = 17.64$$

approx. 17 students initially heard the rumor

- b) How fast is the rumor spreading after 4 days?

$$P(t) = \frac{300}{1+2^{4-t}}$$

$$P'(t) = \frac{(0)(1+2^{4-t}) - (2^{4-t})(\ln 2)(-1)}{(1+2^{4-t})^2}$$

$$P'(t) = \frac{(2^{4-t})(\ln 2)}{(1+2^{4-t})^2}$$

$$P'(4) = 0.17328$$

the rumor spreads @ 0.17328 students/day @ 4 days

5. Seth was asked to derive and fully factor his final answer. His solution is incorrect, but he cannot find his mistake(s). Explain where he made his mistake(s) and correct them.

$$y = 3x^2e^{-2x}$$

$$\frac{dy}{dx} = 6xe^{-2x} + 3x^2e^{-2x}(-2)$$

$$\frac{dy}{dx} = 3e^{-2x}(2x + x^2)$$

should have (-2)

$$y = 3x^2e^{-2x}$$

$$y' = 6x(e^{-2x}) + 3x^2(e^{-2x})(-2)$$

$$= 6x(e^{-2x})(1-x)$$

His mistake is for the exponential function, multiplied to the term he forgot to take the derivative of the exponent.

6. Liam is given the equation $y = \sin x$ and is asked to determine the value of the 50th derivative when $x = \frac{\pi}{2}$.

Explain an alternative strategy for Liam to do this so that he doesn't have to differentiate 50 times. Then, show his solution.

$$y = \sin x$$

$$y' = \cos x$$

$$y'' = -\sin x$$

$$y''' = -\cos x$$

$$y^{(4)} = \sin x$$

$$y^{(5)} = \cos x$$

As can be seen by the first five derivatives, the derivatives become a repeating pattern

due to the fact the $\frac{dy}{dx}$ of $\cos x = -\sin x$

and the $\frac{dy}{dx}$ of $\sin x = \cos x$. Since 50 is

divisible by 4, the 50th derivative should be $y = \cos x$. Plug in $x = \frac{\pi}{2}$ $y = 0$ for the 50th derivative.

Crystal be
The height of a spring over time, in seconds, can be modeled by the function $h(t) = -3 \cos 3t + 3$ over the interval $[0, \frac{3\pi}{2}]$.

- a) At what time(s) does the spring change direction? $[0, \frac{3\pi}{2}]$

$$h(t) = -3 \cos 3t + 3$$

$$h'(t) = 9 \sin 3t$$

$$0 = \sin 3t \quad \text{period} = \frac{2\pi}{k}$$

$$= \frac{2\pi}{3}$$

$$\left. \begin{array}{l} 0 = 3t \quad \pi = 3t \quad 2\pi = 3t \\ t = 0 \quad t = \frac{\pi}{3} \quad t = \frac{2\pi}{3} \\ (0, \frac{\pi}{3}) \quad (\frac{\pi}{3}, \frac{2\pi}{3}) \quad (\frac{2\pi}{3}, \frac{3\pi}{2}) \\ h'(t) \quad (+) \quad (-) \quad (+) \end{array} \right\}$$



changes direction @ $\frac{\pi}{3}s, \frac{2\pi}{3}s$ Add period!

2
(3)

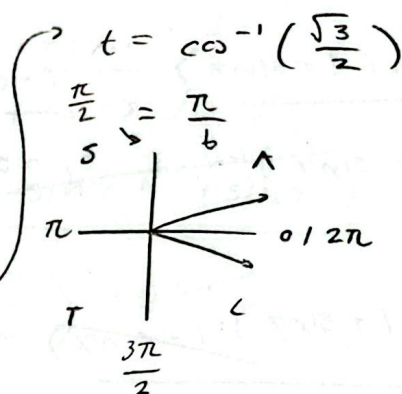
- b) When is the second time the spring's acceleration reaches $\frac{27\sqrt{3}}{2} \text{ m/s}^2$? What is the spring's height at that time? Use exact values.

$$h'(t) = 9 \sin 3t$$

$$h''(t) = 27 \cos 3t$$

$$\frac{27\sqrt{3}}{2} = 27 \cos 3t$$

$$\cos 3t = \frac{\sqrt{3}}{2}$$



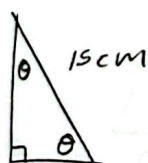
$$t = \frac{\pi}{6}, \frac{11\pi}{6}$$

not in domain

$$h(\frac{\pi}{6}) = 3 \text{ m}$$

the exact time is $\frac{\pi}{6}s$ and it reaches 3 m

8. The hypotenuse of a right triangle is 15 cm in length. Calculate the measures of the unknown angles that will maximize the area of the triangle. You do not need to prove that the critical value is a maximum.



$$a = \frac{lw}{2}$$

$$\sqrt{l^2 + w^2} = 15$$

$$\theta_1 + \theta_2 + 90^\circ = 180^\circ$$

$$\theta_1 + \theta_2 = 90^\circ$$

$$\theta_2 = 90^\circ - \theta_1$$

$$\theta_1 + (90^\circ - \theta_1) = 90^\circ$$

$$f'(x) = x + 90 - x - 90$$

0.5

(3)

4.5

$$\ln(f(x)) = \frac{f'(x)}{f(x)}$$

9. Prove that if $g(x) = \ln\left(\sqrt{\frac{1+\sin x}{1-\sin x}}\right)$ then $g'(x) = \sec x$.

$$\frac{1}{\cos x}$$

$$g(x) = \ln \sqrt{\frac{1+\sin x}{1-\sin x}}$$

$$= \ln \left(\frac{1+\sin x}{1-\sin x} \right)^{\frac{1}{2}}$$

$$g'(x) = \frac{1}{2} \left(\frac{1+\sin x}{1-\sin x} \right)^{-\frac{1}{2}} \left(\frac{\cos x (1-\sin x) - (-\cos x) (1+\sin x)}{(1-\sin x)^2} \right)$$

$$\left(\frac{1+\sin x}{1-\sin x} \right)$$

$$= \frac{\cos x (1-\sin x) + \cos x (1+\sin x)}{2 \left(\frac{1+\sin x}{1-\sin x} \right)^{\frac{1}{2}} (1-\sin x)^2} \times \frac{1-\sin x}{1+\sin x}$$

$$= \frac{\cos x (1-\sin x) + \cos x (1+\sin x)}{2 \left(\frac{1+\sin x}{1-\sin x} \right)^{\frac{1}{2}} (1-\sin x)^2}$$

$$\frac{(1-\sin x)(1+\sin x)}{1-\sin^2 x = \cos x}$$

$$= \frac{\cos x [(1-\sin x) + (1+\sin x)]}{2 \left(\frac{1+\sin x}{1-\sin x} \right)^{\frac{1}{2}} \cos x}$$

$$2 \left(\frac{1+\sin x}{1-\sin x} \right)^{\frac{1}{2}}$$

$$= \frac{1}{\sqrt{\frac{1+\sin x}{1-\sin x}}}$$

$$2 \left(\frac{1+\sin x}{1-\sin x} \right)^{\frac{1}{2}}$$

$$= \frac{1}{\sqrt{\frac{1+\sin x}{1-\sin x}}}$$

$$= \sqrt{\frac{1-\sin x}{1+\sin x}}$$

look @ attached PD